University of Al-Kafeel
College of Dentistry

LEC7

PHARMACOLOGY OF HYPERTENSION

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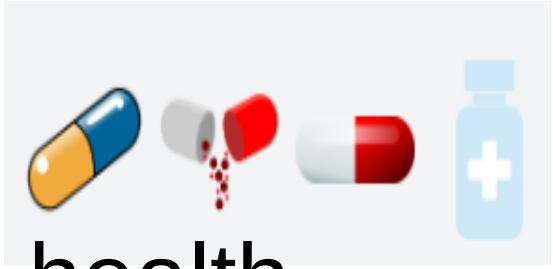
Learning Objectives



By the end of this lecture, you should be able to

- **Define** hypertension and explain its global health significance.
- **Differentiate** between primary (essential) and secondary hypertension.
- **Classify** hypertension based on blood pressure stages according to international guidelines.
- **Describe** the pathophysiological mechanisms and treatment protocol

Definition: Hypertension

- **Hypertension**, or high blood pressure, is a chronic medical condition characterized by sustained elevation of arterial blood pressure above **140/90 mmHg**.
- **Global burden:** It is one of the leading causes of cardiovascular morbidity and mortality worldwide, affecting over **1 billion people**.
- **Importance:** Early detection and management are crucial to prevent complications such as stroke, myocardial infarction, kidney failure, and heart failure

Classification of Hypertension

- Primary (Essential) Hypertension:
  - Accounts for 90–95% of cases.
  - No identifiable single cause; linked to genetic predisposition, lifestyle, and environmental factors.
- Secondary Hypertension:
  - Accounts for 5–10% of cases.
  - Caused by underlying conditions such as renal disease, endocrine disorders (e.g., hyperthyroidism, Cushing syndrome), or medications.

Pathophysiology

- **Vascular resistance**: Increased systemic vascular resistance due to arterial stiffness.
- **Neurohormonal factors**: Overactivation of the sympathetic nervous system and renin-angiotensin-aldosterone system (RAAS).
- **Endothelial dysfunction**: Reduced nitric oxide availability impairs vasodilation.
- **Genetic and environmental interplay**: Salt sensitivity, obesity, and sedentary lifestyle contribute significantly

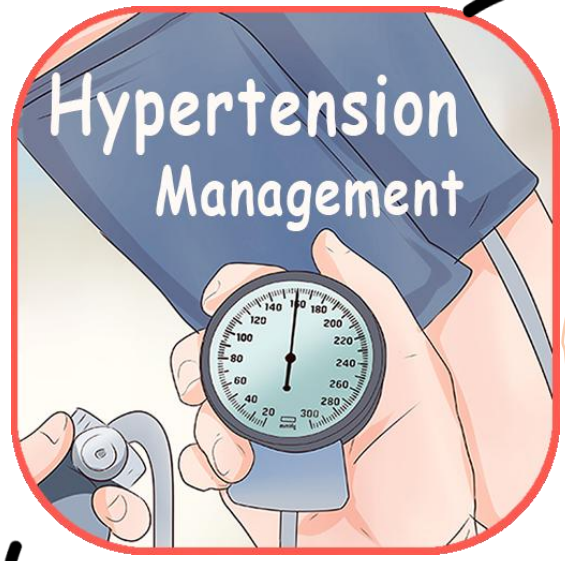
Risk Factors

- Non-modifiable: Age, family history, ethnicity.
- Modifiable:
  - High salt intake
  - Obesity and sedentary lifestyle
  - Smoking and alcohol consumption
  - Stress and poor diet

Complications

- **Cardiovascular:** Coronary artery disease, left ventricular hypertrophy, heart failure.
- **Neurological:** Stroke, transient ischemic attacks, cognitive decline.
- **Renal:** Chronic kidney disease, nephrosclerosis.
- **Ocular:** Hypertensive retinopathy.

Management of hypertension



Non- Pharmacological treatment

Weight loss

Exercise

Salt restriction

Pharmacological treatment

Smoking cessation

DASH plan

Classification of Antihypertensive Drugs

A- Drugs acting on sympathetic NS

1. Beta -blockers
2. Alpha -blockers
3. Mixed alpha and beta blockers
4. Alpha 2-agonists
5. Adrenergic neuron blockers

Beta blockers

- Arterial BP= CO x TPR
- Inhibit renin secretion
- B-blockers are classified into:
  - **Selective:** like atenolol, metoprolol
  - **Non-selective:** like propranolol

Points to be remembered with Beta blockers

They are indicated in hyperkinetic hypertensive patients.

They are contraindicated in :

- Heart block
- Bradycardia
- Asthmatic patients
- Patients with peripheral vascular disease
- Patients with diabetes mellitus

Alpha- blockers

Indications:

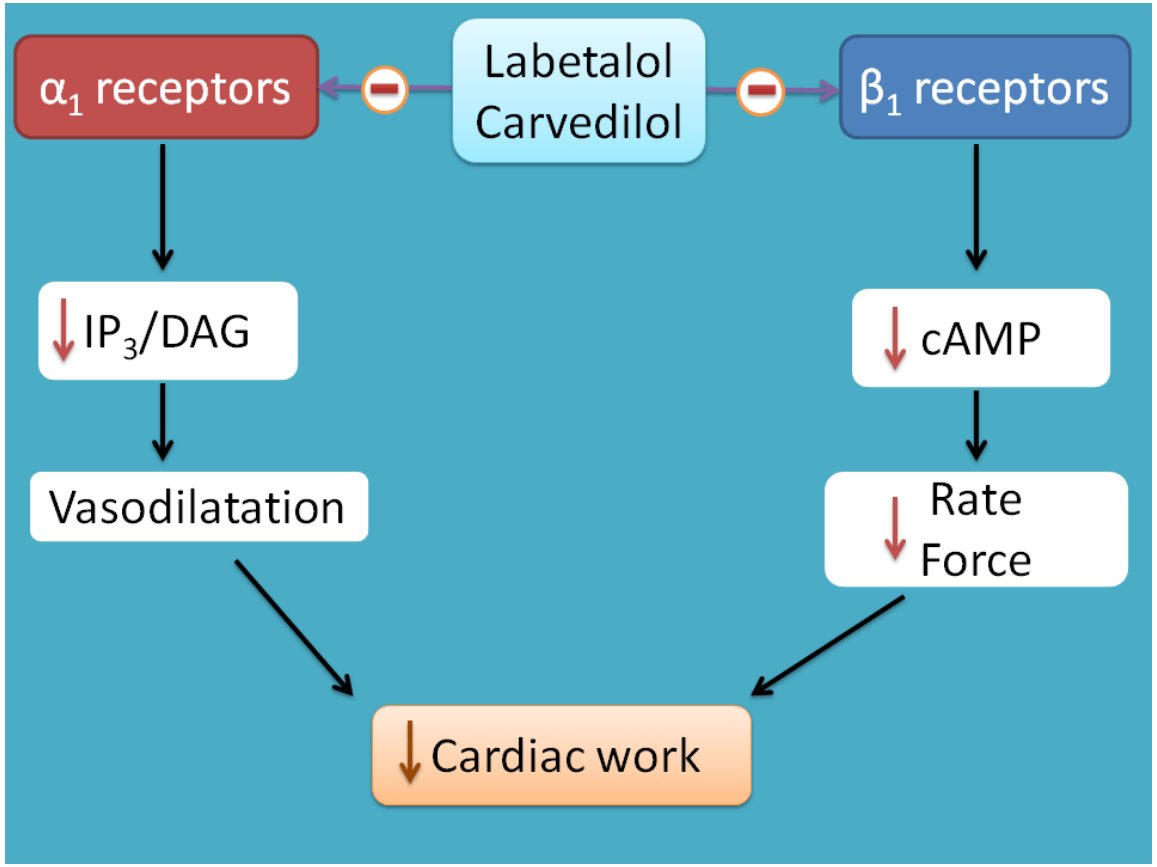
1. Benign prostatic hypertrophy
2. Pheochromocytoma
3. Peripheral vascular disease

Side effects:

First dose phenomenon (Postural hypotension)

Mixed Alpha & Beta- blockers

- **Labetalol :** used in emergency hypertension
- **Carvedilol :** it has additional vasodilator effect



Alpha-2 Agonist

Clonidine

- It is α2-agonist
- Indications:
  - Hypertension (rarely used because of rebound hypertension)
  - Control of menopausal symptoms
  - Treatment of opioid addiction
  - Clonidine test for growth hormone
- Side effects:
  - The most important is rebound hypertension due to sudden withdrawal.

Alpha-2 Agonist

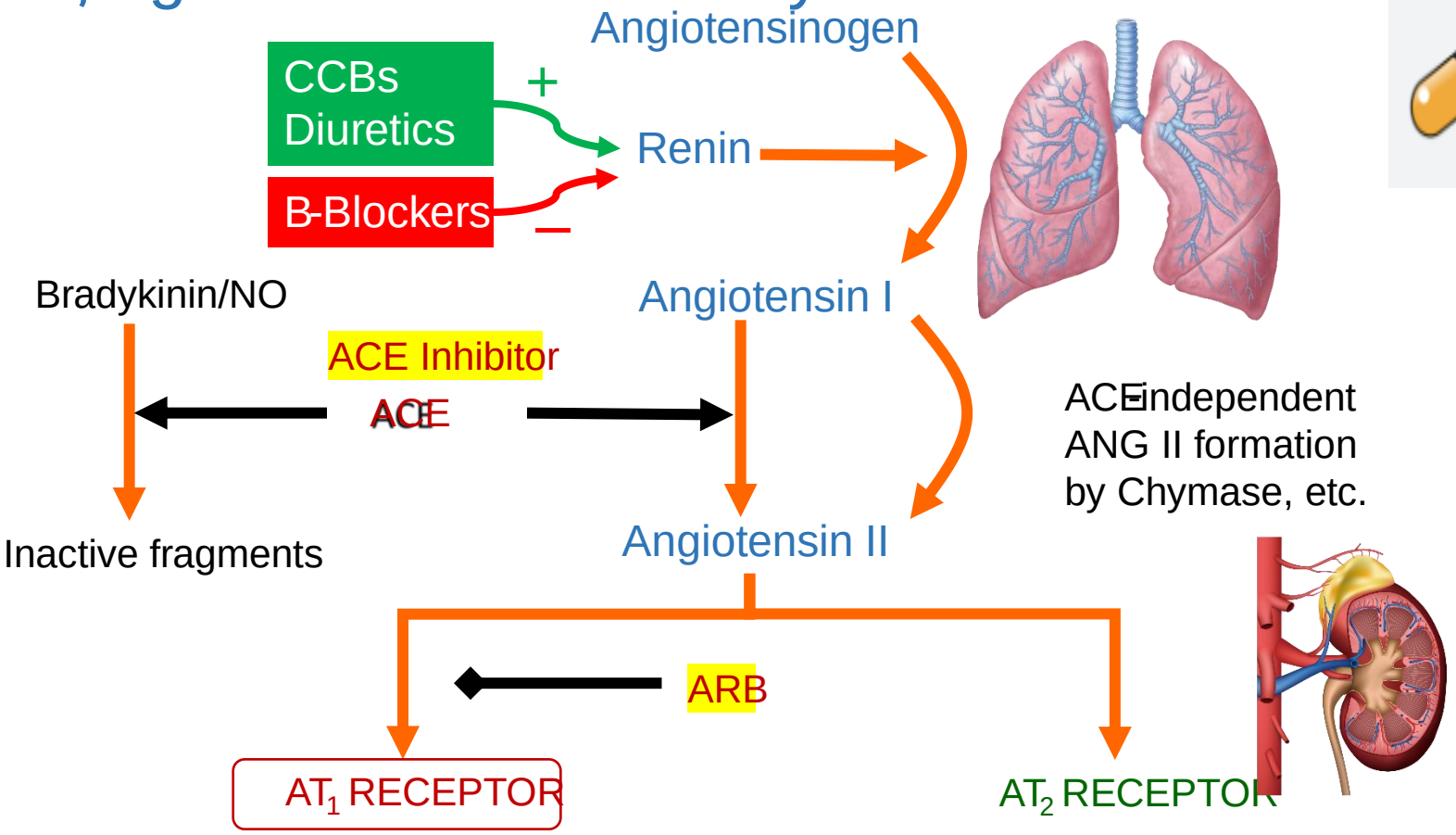
Methyl dopa

- It metabolized into h-methyl noradrenaline
- It is h2-agonist
- It is safe in pregnancy and renal failure
- Side effects are :
  - Headache
  - Thrombocytopenia
  - Involuntary movement
  - Sedation
  - Hemolytic anemia
  - Hepatitis and pancreatitis

Adrenergic neuron blockers

- These such as guanethidine and reserpine
- These drugs cause depletion of noradrenaline
- They cause postural hypotension
- Important side effect :
  - Parkinsonism
  - Depression especially in females leads to suicidal attempts

The renin angiotensin Aldosteron system (RAAS)



Vasoconstriction
Sodium retention
SNS activation
Inflammation

Vasodilation
Natriuresis
Tissue regeneration
Inhibition of inappropriate cell growth

Angiotensin-II

1. Direct vasoconstrictor
2. It acts on two types of receptors
   1. AT-1: mediates vasoconstriction and cardiac hypertrophy
   2. AT-2: mediates antigrowth and antiproliferative
3. It stimulates secretion of aldosterone causing sodium and water retention

ACEIs

ÅIndications:

1. Treatment of hypertension
2. Treatment of all grades of heart failure by balanced vasodilatation

Arterial vasodilatation ω Decrease afterload

Veno-dilatation ω Decrease preload

3. Decrease urinary protein excretion in patients with diabetic nephropathy
4. Post-MI to prevent ventricular remodeling, cardiac dilatation and heart failure.

Captopril

Side effects:

1. Dry cough
2. Angioedema
3. Hyperkalemia
4. Thrombocytopenia and neutropenia
5. Impairment of taste sensation
6. Membranous glomerulonephritis
7. Nephrotic syndrome with proteinuria

Other ACE inhibitors

Enalapril

lisinopril

Angiotensin receptor blocker (ARBs)

- They are called **sartans**
- They **block AT-1** receptor
- **Less cough**

Indications:

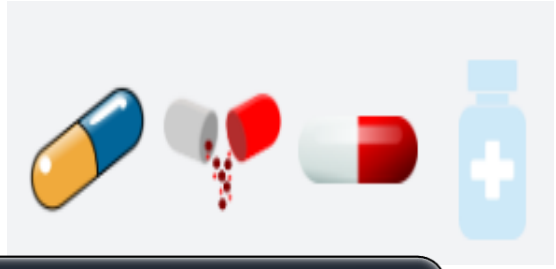
- Hypertension
- Proteinuria due to diabetic nephropathy
- Heart failure
- Post MI

Examples : losartan, valsartan, telmisartan, candesartan, olmesartan, irbisartan, eprosartan.

Renin inhibitors

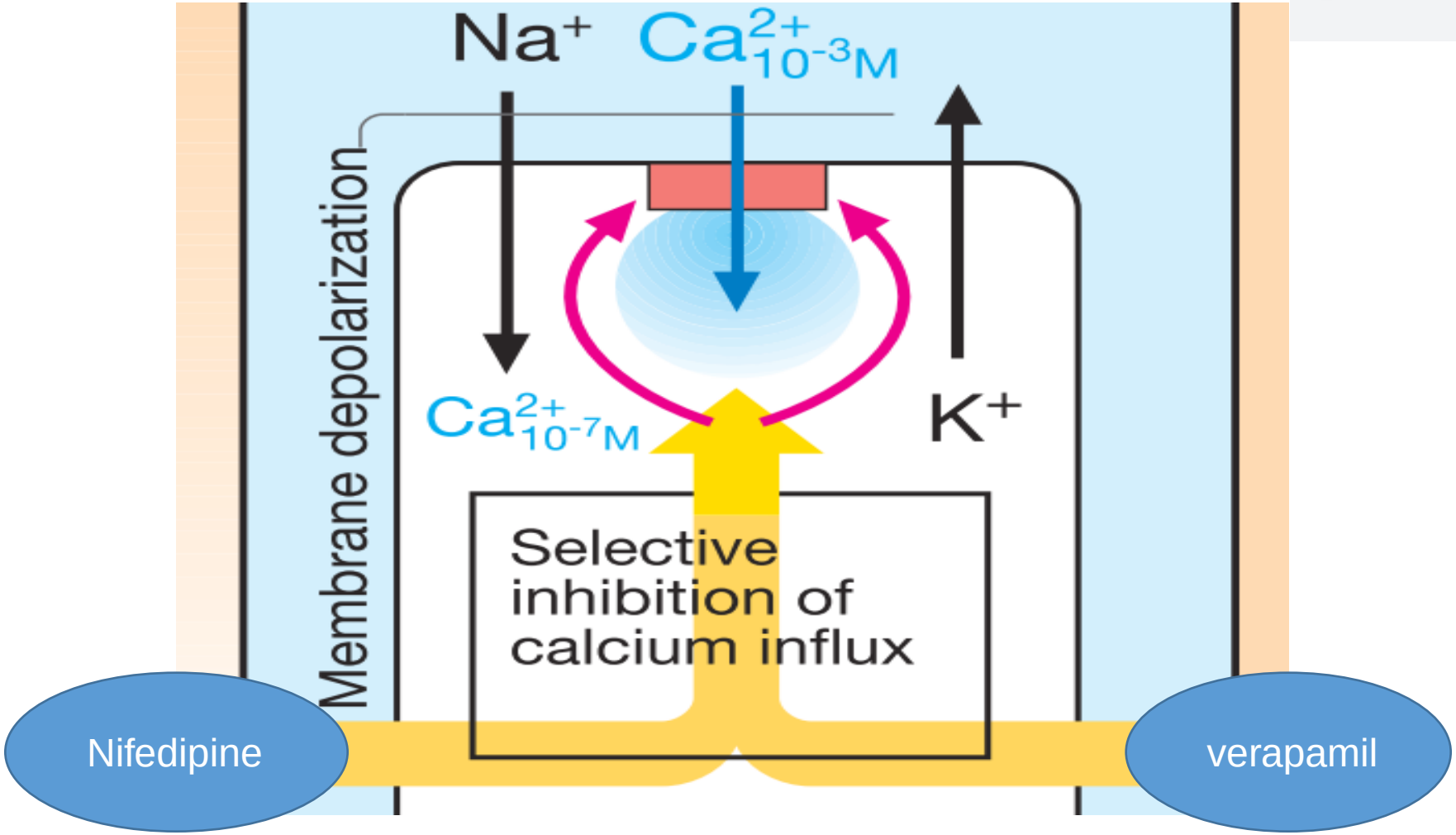
1. Alpha -methyl -dopa
2. Beta -blockers
3. Aliskirin, enalkirin

Direct acting vasodilators



- Diazoxide
- Hydralazine
- Minoxidil
- Sodium nitoprusside

Calcium Channel Blocker



Nifedipine

- Vasodilator arteries > veins
- Negative inotropic effect
- Has no antiarrhythmic effect
- Contraindicated in heart failure
- **Side effect:**
  - Reflex tachycardia associated with increased CV diseases
  - Headache due to vasodilatation
  - Gum hypertrophy
  - Ankle swelling due to relaxation of the precapillary sphincter more than the post capillary sphincter
  - Facial flushing

CCBs

- **Amlodipine:**
  - Long acting
  - The only CCB **safe in heart failure**
- **Nicardipine :**
  - It is used in hypertensive emergency
- **Diltiazem:**
  - **It is less cardiac depressant than verapamil**
  - **Indicated in treatment of hypertension and angina**
  - Side effects:
    - **Peripheral neuropathy**
    - **Changes in liver enzymes**
    - **Myocardial depressant effect**
    - **Depression and headache on continuous use.**

Verapamil

- Affects **cardiac** and **vascular** calcium channels.
- ↓myocardial contractility ↑refractory period of AV node so delay conduction at AV node
- Indications:
  - Hypertension
  - Angina pectoris
  - Supraventricular tachycardia
- Contraindications:
  - Wolff -Parkinson -White syndrome (WPW)
  - Should not be combined with BB
  - Heart block
- Side effects

Hypertensive Emergencies

- **High blood pressure associated with:**
  1. Dissecting aortic aneurysm
  2. Hypertensive encephalopathy
  3. Acute left sided heart failure (acute pulmonary oedema)
- **Treatment:**
  - i.v. diazoxide
  - i.v. sodium nitroprusside
  - i.v. hydralazine
  - i.v. infusion of nitroglycerine
  - i.v. labetalol
  - Sublingual nifedipine

Treatment of hypertension during pregnancy

1. Methyl -dopa → **safe**
2. Hydralazine → **safe**
3. CCBs → **safe as** nifedipine
4. BBs → **unsafe** neonatal bradycardia, neonatal hypoglycemia except for Labetalol
5. Diuretics → **unsafe** because of placental calcification and placental insufficiency
6. ACEIs and ARBs → **unsafe** because of fetal renal dysgenesis and renal failure.

References

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